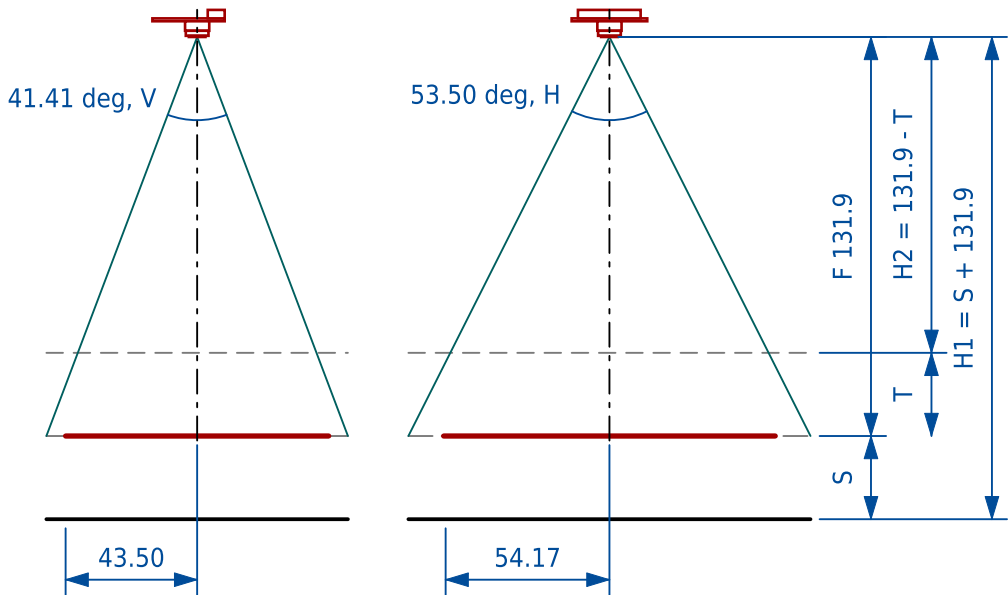
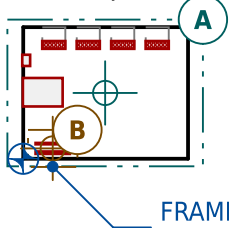


END ELEVATION

FRONT ELEVATION



PLAN, 1:5



NOTES

1. First angle; the end elevation is seen from the right. One camera: a Raspberry Pi Camera Module v1.3, stock 65 deg lens, looking straight down, reading the whole sensor.

NOTES (continued)

2. Heights are to the lens FACE. The entrance pupil the optics are worked to is ASSUMED behind it by at most the lens's 5.20 mm, so the picture is up to 9.2% larger, never smaller, and the board and its LEDs no nearer than the close limit.
3. F is, per frame, the higher of two heights: the one that just frames it with the lens focused there, frame A  $128.3 + f\ 3.60 = 131.9$ , frame B  $32.4 + f\ 3.60 = 36.0$  (DERIVED, thin lens), and the lens's close limit unscrewed, 60 mm: "About the closest you can get this way is about 6 cm." (a Raspberry Pi forum user, 2013; REPORTED). The forum does not say whether its "about 6 cm" is from the lens face; it is taken as from the face, note 2's convention.
4.  $H1 = S + F$ , frame A  $S + 131.9$ , frame B  $S + 60.0$ , where S is what the Arty stands on, under its rubber feet, to its top face: a foot and the board: MEASURE, not published.
5.  $H2 = F - T$ , the clearance, frame A  $131.9 - T$ , frame B  $60.0 - T$ , where T is the Arty's top face to the top of its tallest part: MEASURE, not published.
6. HEADROOM: a part standing on the board, anywhere in its envelope stays in frame A's picture, the lens refocused to F, up to 13.6 mm above the board's top face at the envelope's corner, the worst case; where J9 stands, up to 20.9 (DERIVED). If T's part stands higher than its figure, raise the camera.

NOTES (continued)

7. REFOCUS THE LENS TO F, 131.9 mm from the lens face to the board and its LEDs for frame A, 60.0 for frame B. As sold it is set far -- "Approx 1 m to infinity" (Raspberry Pi; infinity is their sign), "from about 0.5m to infinity" (raspi.tv) -- and at frame A's F a point spreads to 23 px (DERIVED). raspi.tv adds a +2D lens in front, "focus at about 25cm", too far for F; a reader's comment there: "You CAN change the focus of the stock lens on the pi camera. It is tricky but can be done." Unscrew it until the board and its LEDs are sharp.
8. A frame is the sensor's own 4:3 round its target plus 5.00 mm all round, turned whichever way needs the lower camera; the table gives which, and where the lens axis goes over each.
9. Frame A, Whole board: The assembled envelope: the outline together with the connectors and Pmod bodies that overhang it.
10. Frame B, LD0-LD7: The four tri-colour LEDs LD0-LD3 and the four single LD4-LD7, in two rows on a 7.00 mm pitch.
11. Which LED row is which is not named by any Digilent source; FPGA-ARTY-A7 takes the row nearest the edge as the tri-colour LD0-LD3 by package size. Either way both rows are inside the frame, so the framing does not turn on it.
12. S: the A7 drawing has four corner circles, 10.0 mm across scaled from its plot, taken as the rubber feet, and no height for them or the board. Digilent's 3D model of the Arty Rev C, not the A7, has a 1.50 board on feet 3.70 tall, 5.20 in all, but its feet are 8.70 across and it states no tolerance. Measure S on the A7, from what it stands on to its top face.

LENS FACE HEIGHTS, mm: BOARD IN VIEW AND IN FOCUS

|    | THE LENS FACE ABOVE                          | FRAME A     | FRAME B    | TERM       |
|----|----------------------------------------------|-------------|------------|------------|
| H1 | what the Arty stands on, under its feet      | $S + 131.9$ | $S + 60.0$ | S: MEASURE |
| H2 | the highest point on the board, as clearance | $131.9 - T$ | $60.0 - T$ | T: MEASURE |
| F  | the Arty's top face                          | 131.9       | 60.0       | DERIVED    |

FRAMES, AND WHERE THE LENS AXIS GOES, mm

|   | FRAME       | RECTANGLE      | LONG | AXIS X | AXIS Y |
|---|-------------|----------------|------|--------|--------|
| A | Whole board | 129.33 x 97.00 | X    | 54.17  | 43.50  |
| B | LD0-LD7     | 32.68 x 24.51  | X    | 19.50  | 6.83   |

LEGEND

- Subject outline
- LED, display or connector
- Adjacent part or connector body
- Frame A's target, on the plane F is measured from
- The 65 deg lens's field of view, 53.50 x 41.41, to its picture's edge
- Frame A, the rectangle the picture must cover
- Frame B, the rectangle the picture must cover
- What the Arty stands on, S below the board's top face: NOT TO SCALE, to measure
- The highest point on the board, T above the board's top face: NOT TO SCALE, to measure
- Dimension, extension and leader

NOTES (continued)

13. T: the A7 drawing gives no part a height. The Rev C model's tallest part is the Ethernet jack J9's shield, 14.80 above the board's top face, and the A7 drawing puts J9 in the same place: more than the 13.6 mm headroom at the envelope's corner, less than the 20.9 where J9 stands. Measure T on the A7, from its top face to the top of its tallest part, and which part.

SOURCES

1. Board geometry: fpga/boards.py - Outline, Pmod hosts, connectors and LED rows; see FPGA-ARTY-A7.
2. 3D model, Arty Rev C: [https://digilent.com/reference/\\_media/reference/programmable-logic/artys/artys\\_rev\\_c\\_cad.zip](https://digilent.com/reference/_media/reference/programmable-logic/artys/artys_rev_c_cad.zip) - Digilent, Arty Rev C.stp, 20 June 2016: not the A7. Its heights are quoted in the notes; no height on this sheet is set from it.
3. raspi.tv, 25 May 2013: <https://web.archive.org/web/20260212013652/https://raspi.tv/2013/adapt-your-raspberry-pi-camera-for-close-up-use>
4. Lens figures and focus: RPICAM-LENS - Every lens figure on this sheet, declared and derived, with the vendor pages it is quoted from.

CAMERA OVER THE ARTY A7

Camera Module v1.3, stock 65 degree lens: lens face heights

|                             |                 |                                                                        |                                |             |  |                          |                                                                                                     |
|-----------------------------|-----------------|------------------------------------------------------------------------|--------------------------------|-------------|--|--------------------------|-----------------------------------------------------------------------------------------------------|
| DRAWN<br>generated          |                 | VERSION<br>v0.0-606-gd307ee4                                           |                                | UNITS<br>mm |  | SCALE<br>1:2.5, PLAN 1:5 |                                                                                                     |
| SIZE<br>A3                  | SHEET<br>1 OF 1 |                                                                        | DRAWING NO<br>RPICAM-OVER-ARTY |             |  |                          | REV<br>A                                                                                            |
| SUBJECT<br>Digilent Arty A7 |                 | GENERAL TOLERANCE<br>DXF +/-0.20, plot bodies +/-0.30, heights DERIVED |                                |             |  |                          | PROJECTION<br> |